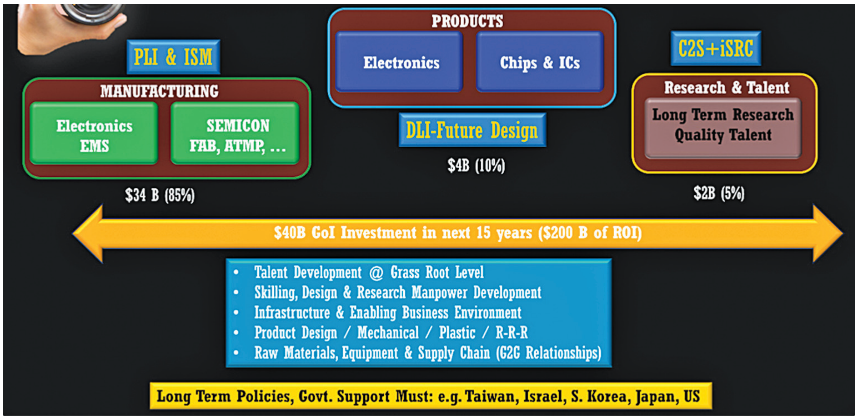

Fig. 5: Self-reliance vision for electronics and semiconductors

Electronics products provide a market for semiconductor chips, and the need of the hour is to provide similar incentives as DLI for domestic electronics product companies and startups to create a strong design and innovation ecosystem in this very critical sector. Ideally, 10% of money allocated for electronics and semiconductor manufacturing under the PLI and other schemes should be allocated as incentives for domestic electronics and semiconductor product companies and startups. Considering $10 billion has been allocated for electronics manufacturing under the PLI scheme, allocating $1 billion for design-linked incentives for electronics products and systems would be a game changer and help grow sustainable and high-value-added electronics manufacturing for domestic consumption and exports.

Creating a conducive environment for hardware startups in India is vital for achieving self-reliance and global competitiveness in the electronics and semiconductor industry. By re-balancing incentives, raising awareness, and providing mentorship and financial support, India can nurture a new generation of successful hardware startups and drive innovation in this critical sector. Leveraging organisations like VLSI Society of India (VSI), India Electronics and Semiconductor Association (IESA), ELCINA, STPI, and hundreds of incubators can be instrumental in bringing together stakeholders, fostering collaboration, and providing valuable support to the hardware startups for both semiconductor and electronics products.

Vision for self-reliance in electronics and semiconductors

For India to become a self-reliant and globally competitive trusted partner in electronics and semiconductor supply chain, a balanced approach focusing on products, manufacturing, and talent development and research is needed. When we examine these three essential pillars, it is evident that significant strides have been made in manufacturing, especially in electronics manufacturing services (EMS) and contract manufacturing.

The Indian government has introduced schemes such as the Production-Linked Incentive (PLI) scheme for electronics manufacturing with almost $20 billion in incentives and the India Semiconductor Mission $10 billion for semiconductors for much needed semiconductor ecosystem. Electronics manufacturing, with strong support from PLI and other incentive schemes, has surpassed the $100 billion mark with more than $20 billion in exports and is expected to grow to $300 billion and $100 billion just in exports.

For semiconductor manufacturing, which is strategically critical, India Semiconductor Policy of 2022 has provided a good impetus. The government of India, many state governments, industry, academia, and many global Indians have put in outstanding efforts over the last 18-24 months and many projects, including that of Micron and Tata Group in semiconductors, are in various stages of planning, government approval, and execution.

India can say, "Mere paas MA hai," fast-growing 'MA'rket, globally competitive 'MA' nufacturing, and the large pool of talented 'MA'npower. The long-term vision's third and most critical pillar is 'P'roducts. The Semiconductor DLI scheme is a good start in this direction. However, significant work is still needed for domestic electronics and semiconductor product design and brand creation to serve the solid domestic and global market, utilise and make the manufacturing ecosystem sustainable and leverage strong manpower.

India's strategy has been manufacturing, and incentives are more tilted towards manufacturing. Providing incentives for product development equivalent to 10% of manufacturing incentives, or approximately $2 billion, will help create thousands of product companies, ODMs, OEMs, brands, and unicorns in the electronics and semiconductor field. Taking advantage of India's strengths, a balanced focus on products, manufacturing, talent, R&D, and proportionate incentives will propel India to become 'An Electronics and Semiconductor Product Nation.' EFY

This article is based on a tech talk session at EFY Expo 2023, Delhi, by Dr Satya Gupta, President, VLSI Society of India, Intel Veteran, Founder Open-Silicon, EPIC Foundation and India Electronics and Semiconductor Association. It has been transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Journalist at EFY.